

Problem

Design a problem to help other students better understand how to find the inverse Laplace transform.

Step-by-step solution

Step 1 of 3

Design a problem to help other students better understand how to find the inverse Laplace transform.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

Find the inverse Laplace transform of the following functions:

$$(a) \quad F(s) = \frac{20(s+2)}{s(s^2+6s+25)}$$

Step 2 of 3

$$(b) \quad P(s) = \frac{6s^2+36s+20}{(s+1)(s+2)(s+3)}$$

Solution

$$(a) \quad F(s) = \frac{20(s+2)}{s(s^2+6s+25)} = \frac{A}{s} + \frac{Bs+C}{s^2+6s+25}$$

$$20(s+2) = A(s^2+6s+25) + Bs^2 + Cs$$

Equating components,

$$s^2: \quad 0 = A + B \text{ or } B = -A$$

$$s: \quad 20 = 6A + C$$

$$\text{constant:} \quad 40 - 25A \text{ or } A = 8/5, B = -8/5, C = 20 - 6A = 52/5$$

$$F(s) = \frac{8}{5s} + \frac{-\frac{8}{5}s + \frac{52}{5}}{(s+3)^2 + 4^2} = \frac{8}{5s} + \frac{-\frac{8}{5}(s+3) + \frac{24}{5} + \frac{52}{5}}{(s+3)^2 + 4^2}$$

$$\underline{f(t) = \frac{8}{5}u(t) - \frac{8}{5}e^{-3t}\cos 4t + \frac{19}{5}e^{-3t}\sin 4t}$$

Step 3 of 3

$$(b) \quad P(s) = \frac{6s^2 + 36s + 20}{(s+1)(s+2)(s+3)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3}$$

$$A = \frac{6 - 36 + 20}{(-1+2)(-1+3)} = -5$$

$$B = \frac{24 - 72 + 20}{(-1)(1)} = 28$$

$$C = \frac{54 - 108 + 20}{(-2)(-1)} = -17$$

$$P(s) = \frac{-5}{s+1} + \frac{28}{s+2} - \frac{17}{s+3}$$

$$p(t) = \underline{(-5e^{-t} + 28e^{-2t} - 17e^{-3t})u(t)}$$